DIFFERENTIAL AMPLIFIERS AND CURRENT MIRRORS

Current Mirrors

→ $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(\frac{V_{DD}R_1}{R_1+R_2} - V_{Th}\right)$ in a voltage divider bias circuit

I_D depends on μ_n, V_{DD} & V_{Th} if temperature changes

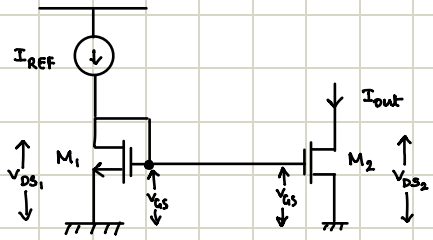
To overcome this, we use Current Mirror!

→ We assume some I_{REF} which is constant current source (Golden Reference)

and we copy this current

→ Normally, if the cause is V_{GS}, the effect is I_D (Change in V_{GS} causes change in I_D)

But now, the cause is I_D, the effect is V_{GS} (Change in I_D causes change in V_{GS})



Basic Current Mirror

$V_{GS_1} = V_{GS_2}$

$I_{out} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{Th})^2$

$I_{REF} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{Th})^2$

$I_{out} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} \cdot I_{REF}$ ⇒ I_{out} only depends on I_{REF} & $\left(\frac{W}{L}\right)$ ratios

→ But this doesn't guarantee that $I_{out} = \frac{(W/L)_2}{(W/L)_1} \cdot I_{REF}$ because, I_D depends on V_{DS} as well.

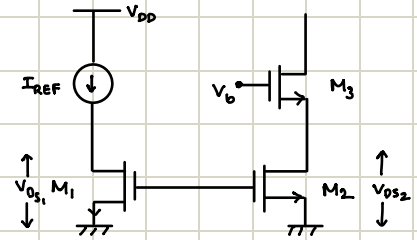
We only are sure $V_{GS1} = V_{GS_2}$ but not $V_{DS_1} = V_{DS_2}$

and hence use only for simple applications.

→ To overcome this issue, we use Cascode Current Mirror

Cascode Current Mirrors

→



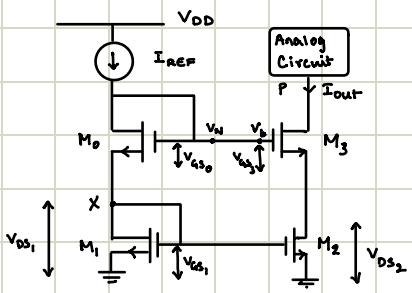
$V_b - V_{GS_3} = V_{DS_2}$

$= V_{DS_1}$ (V_{DS_1} should be equal to V_{DS_2})

$V_D = V_{GS3} + V_{DS1}$

$= V_{GS_3} + V_{GS1}$ ($V_{GS1} = V_{DS1}$)

→



Cascode Current Mirror

$V_N = V_{GS_0} + V_x$

$V_y = V_{DS_2}$

$V_b = V_{GS_3} + V_y$

Now we should make $V_N = V_b$

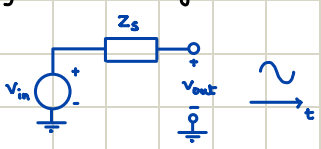
$V_{GS_0} + V_x = V_{GS_3} + V_y$

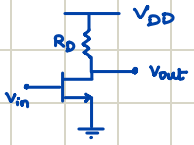
$V_x = V_y$ ($V_{GS_0} = V_{GS_3}$)

Then $V_{DS_1} = V_{DS_2}$ ($V_x = V_y$)

Single Ended and Differential Operation

→ "Single ended signal" is one that is measured wrt fixed potential (usually ground)

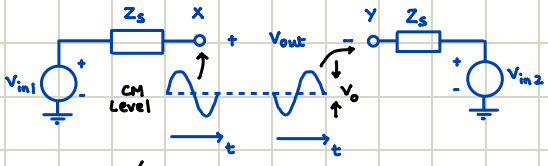




ex: CS Amplifier

⇒ If each single ended output has a peak amplitude voltage of V_o, then single ended peak to peak voltage is $2V_o$

→ "Differential signal" is one that is measured b/w 2 nodes that have equal & opp. signal excursions around fixed potential ex: Operational Amplifier



↳ V_{in1} & V_{in2} are out of phase

$V_{out} = V_x - V_y$

$V_{in1} = V_o \sin\omega t + V_{CM}$

$V_{in2} = -V_o \sin\omega t + V_{CM}$

$V_{out} = V_{in1} - V_{in2}$

$= 2V_o \sin\omega t$

$V_{out} = 2(2V_o \sin\omega t) = 4V_o \sin\omega t$

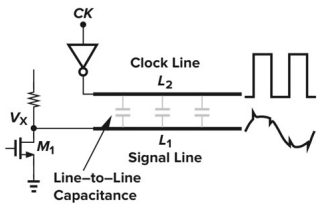
Max $V_{out} = 4V_o$

→ If each single-ended output has peak amplitude of V_o
Then, single-ended peak-to-peak swing is $2V_o$
& differential peak-to-peak swing is $4V_o$

→ The "center" potential in differential signaling is called common mode
↳ Bias value

Advantages of Differential Operation

→ Higher immunity to environmental noise
So for example, take a system where 2 adjascent lines in a circuit carry small sensitive signal & large clock waveform, & due to capacitive coupling b/w lines, transition on L_2 corrupts signal on L_1



V_{in1} & V_{in2} vary by V_{noise}

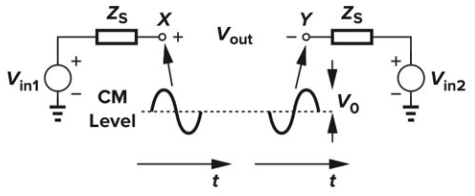
$V_{in1} = V_{in1} + V_{noise}$

$V_{in2} = V_{in2} + V_{noise}$

$V_d = V_{in1} + V_{noise} - (V_{in2} + V_{noise})$

$V_d = V_{in1} - V_{in2}$

→ But in another case, sensitive signal is distributed as 2 equal & opposite phases so if clock line is placed at midway b/w the two, transitions disturb differential phases by equal amounts, leaving difference intact called CM rejection
Even though common-mode level of 2 phases is disturbed, differential output isn't corrupted



→ CM rejection also occurs with noisy supply voltages

In CS Stage, $V_{DD} = V_{DD} \pm \Delta V$ (→ V_{DD} varies by ΔV)

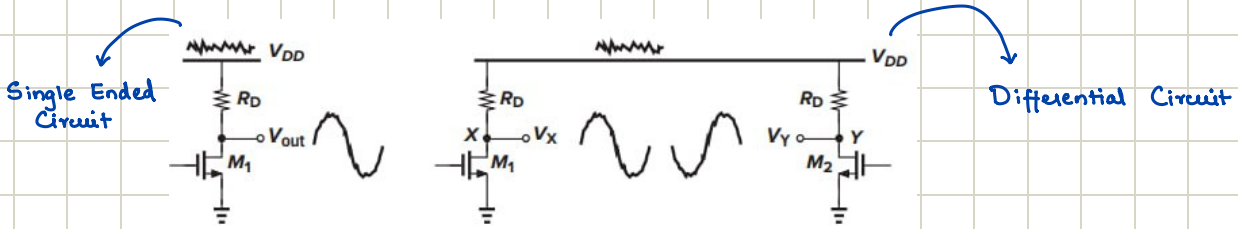
$V_{out} = (V_{DD} \pm \Delta V) - I_D R_D$

So, output is quite susceptible to noise on V_{DD}

If circuit is symmetric, V_{DD} affects V_X & V_Y

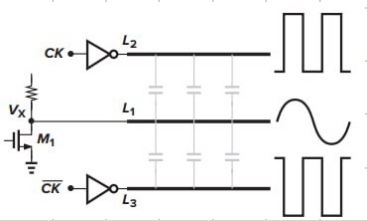
but, $V_X - V_Y = V_{out}$ which doesn't change

So, Differential circuit's signal is more robust in dealing with supply noise



→ In Conclusion, we can employ differential path for sensitive signals (→ victims) & noisy signals (→ aggressors)

Suppose the clock signal is distributed in differential form on 2 lines, then with perfect symmetry, components coupled from CK & $\overline{CK}$ to the signal line cancel each other



→ Differentiable signaling increase max achievable voltage swings

The max output swing at X or Y = $V_{DD} - (V_{GS} - V_{TH})$

But peak to peak swing for $V_X - V_Y$ = $2[V_{DD} - (V_{GS} - V_{TH})]$

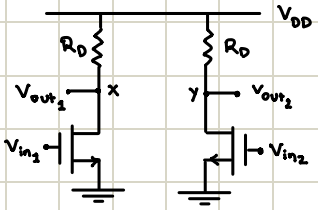
→ Simpler Biasing

→ Higher Linearity

Disadvantage of Differential Operation

→ Occupies Twice the area of single-ended alternatives

Basic Differential Pair



→ V_{in1} , V_{in2} ⇒ Differential Input

V_{out1} , V_{out2} ⇒ Differential Output

$V_{in,CM}$ ⇒ Bias Potential

$V_{out,CM} > V_{in,CM}$

→ $A_v = \frac{V_{out}}{V_{in}} \Rightarrow V_{out} = A_v V_{in}$

→ $V_{out_1} = -g_m R_{D_1} V_{in_1}$ & $V_{out_2} = -g_m R_{D_2} V_{in_2}$, $R_{D_1} = R_{D_2} = R_D$

$V_o = V_{out_2} - V_{out_1}$

$= -g_m R_D (V_{in_2} - V_{in_1})$

$\Rightarrow \boxed{\frac{V_o}{V_d} = -g_m R_D}$ $(V_d = V_{in_2} - V_{in_1})$

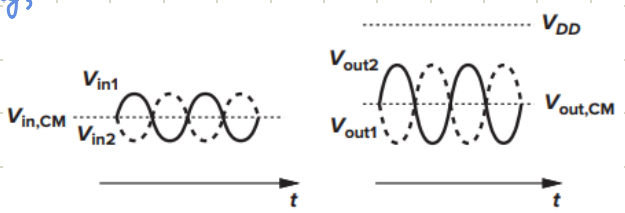
→ Circuit amplifies difference of input signal

Q. What happens if V_{in1} & V_{in2} don't have a well defined CM dc level?

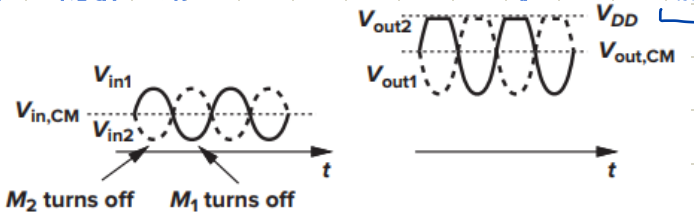
A. If any variation in $V_{in,CM}$ occurs, I_D changes causing g_m and A_v to change

This causes $V_{out,CM}$ to change

Ideally,

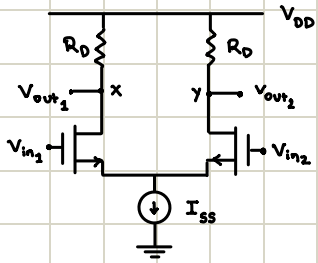


BUT, if input CM level is excessively low, then minimum value of V_{in1} & V_{in2} may turn off M1 & M2 leading to clipping of corresponding output



Avoiding clipping of output (Current Source)

→ At the tail end of basic differential pair, we include a current source I_{SS}



$I_{SS} = I_{D_1} + I_{D_2}$

$R_{D_1} = R_{D_2} = R_D$

$V_{in1} = V_{in2}$

$I_{SS} = 2 I_D$

$V_{out} = V_{DD} - I_D R_D \Rightarrow \boxed{V_{out} = V_{DD} - \frac{I_{SS} R_D}{2}}$

Output depends on I_{SS} rather than I_D

Qualitative Analysis / Behaviour / Operation of Differential Amplifiers

→ Assume $V_{in1} - V_{in2}$ varies from $-\infty$ to ∞

Case 1 : If V_{in1} is more negative than V_{in2}, M_1 is OFF, M_2 is ON

Then $I_{D_2} = I_{SS}$

$V_{out1} = V_{DD}$

$V_{out2} = V_{DD} - I_D R_D = V_{DD} - I_{SS} R_D$

Case 2 : As V_{in1} is brought closer to V_{in2},

M1 gradually turns ON

Partial amount of I_{SS} is drawn from R_{D1} & V_{out1} decreases

$I_{D_1} \uparrow \quad I_{D_2} \downarrow$

$I_{D_1} + I_{D_2} = I_{SS} \Rightarrow V_{out1} \downarrow , \; V_{out2} \uparrow$

Case 3 : When $V_{in1} = V_{in2}$, Both M1 & M2 are ON

$V_{out1} = V_{out2} = V_{DD} - R_D \frac{I_{SS}}{2}$

Case 4 : When $V_{in1} > V_{in2}$,

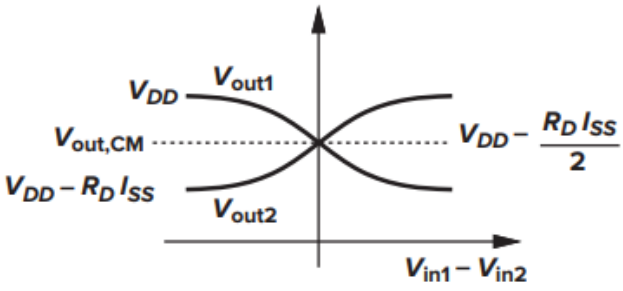
M1 is ON, M2 is OFF

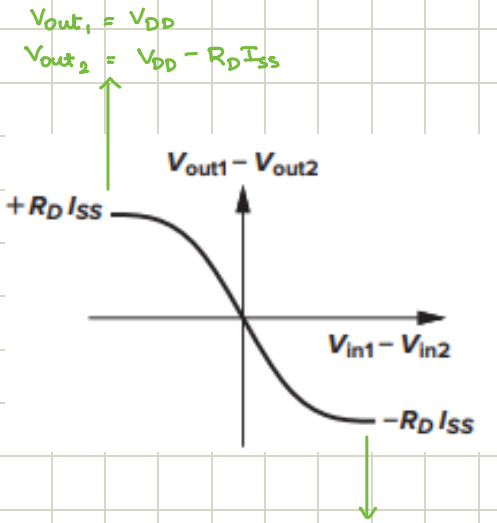
Then, $I_{D_1} = I_{SS}$ & $I_{D_2} = 0$

$V_{out1} = V_{DD} - I_{SS} R_D$

$V_{out2} = V_{DD}$

I/o Char. of Differential Amplifier



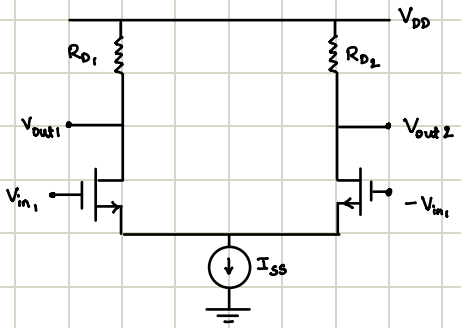


→ Maximum & Minimum values at output are well defined & independant of input CM level

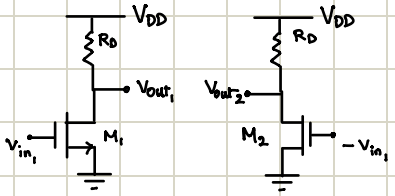
→ Small signal gain (slope of $V_{out1} - V_{out2}$ v/s $V_{in1} - V_{in2}$) is max for $V_{in1} = V_{in2}$ but reduces to zero as $|V_{in1} - V_{in2}|$ increases

Small signal analysis of Differential Amplifier

→ Here we use "Half Circuit Concept"



Small Signal Model ⇒



$$V_{out_1} = -g_m R_d (V_{in_1})$$

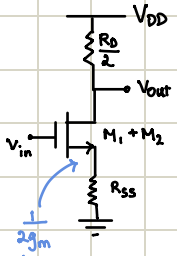
$$V_{out_2} = -g_m R_d (-V_{in_1})$$

$$V_{out_1} - V_{out_2} = -g_m R_D (V_{in_1}) - \left(-g_m R_d (-V_{in_1})\right)$$

$$= -2 g_m R_d V_{in_1}$$

$$\boxed{-g_m R_d = \frac{V_{out_1} - V_{out_2}}{-2V_{in_1}}}$$

Common - Mode Response



2 of them are ||el

$$A_{v,CM} = \frac{V_{out}}{V_{in,CM}} = \frac{-\frac{R_D}{2}}{\frac{1}{2g_m} + R_{SS}}$$

↳ Resistance at drain

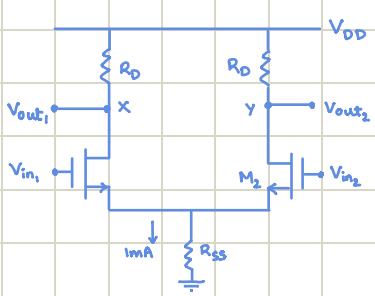
↳ Resistance at source

$$\boxed{A_{v,CM} = \frac{-g_m R_D}{1 + 2 g_m R_{SS}}}$$

where g_m is transconductance of each of M_1 & M_2 AND $\lambda = \gamma = 0$

Here, it is a symmetric circuit, input CM variations disturb the bias points altering small-signal gain and limiting output voltage swings

Q. The circuit of Fig. 4.30 uses a resistor rather than a current source to define a tail current of 1 mA. Assume that $(W/L)_{1,2} = 25/0.5$, $\mu_n C_{ox} = 50\ \mu A/V^2$, $V_{TH} = 0.6$ V, $\lambda = \gamma = 0$, and $V_{DD} = 3$ V.
(a) What is the required input CM voltage for which R_{SS} sustains 0.5 V?
(b) Calculate R_D for a differential gain of 5.
(c) What happens at the output if the input CM level is 50 mV higher than the value calculated in (a)?



A. a) $V_{in,CM} = ?$

$I_{SS} = 1mA \Rightarrow I_{D_1} = I_{D_1} = \frac{I_{SS}}{2} = 0.5mA$

$I_D = \mu_n C_{ox} \frac{W}{L} \frac{(V_{GS} - V_{TH})^2}{2} \Rightarrow V_{GS} = \sqrt{\frac{2 I_D}{\mu_n C_{ox} \frac{W}{L}}} + V_{TH} = V_{GS_1} = V_{GS_2}$

$= \sqrt{\frac{2 \times 0.5 \times 10^{-3}}{50\mu \cdot \frac{25}{0.5}}} + 0.6 = 1.23V$

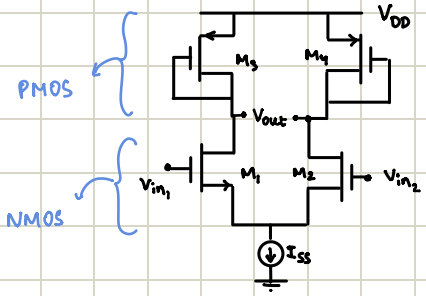
$V_{in,CM} = V_{GS_1} + 0.5 = 1.23 + 0.5 = 1.73V$

b) $g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_{D_1}} = \sqrt{2 \times 50\mu \times \frac{25}{0.5} \times 0.5 \times 10^{-3}} = 1.58\ mS$; $A_v = g_m R_D \Rightarrow R_D = \frac{5}{1.58 \times 10^{-3}} = 3162\ \Omega$

c) $V_{in,CM}$ increases by 50mV $\Rightarrow |\Delta V_{x,y}| = \Delta V_{in}$

Differential Pair with MOS Loads

i) Diode Connected



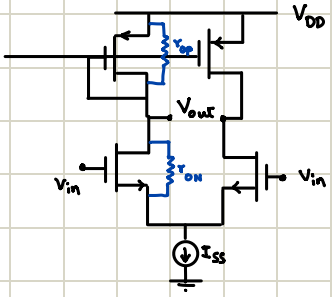
$$A_V = \frac{g_{m1}}{g_{m2} + g_{mb2}}$$

$$g_{mb2} = 0$$

$$A_V = \frac{g_{m_1}}{g_{m_2}} = \frac{-g_{m_N}}{g_{m_P}}$$

$$A_V = \sqrt{\frac{\mu_n \left(\frac{W}{L}\right)_N}{\mu_p \left(\frac{W}{L}\right)_P}}$$

ii) Current Source Load

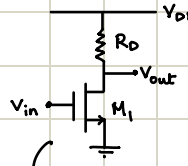


$$A_V = -g_m (R_D \| r_o)$$

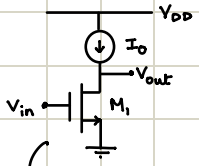
$$= -g_m (r_{oN} \| r_{op})$$

Differential Amplifier with Passive Load

→

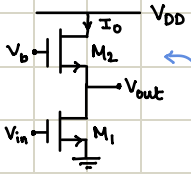


Common Source Amplifier



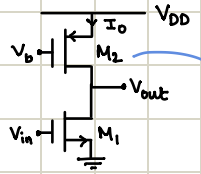
Common Source Amplifier with large A_v

=



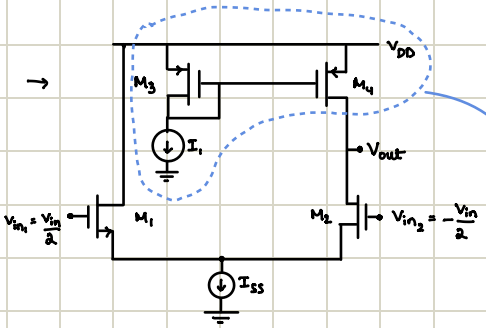
But shows Body Effect

→ To avoid it, we can use PMOS



Now no body effect

→



$$V_d = V_{in}$$

$$V_{in_1} - V_{in_2} = V_{in} \Rightarrow \frac{V_{in}}{2} - \left(-\frac{V_{in}}{2}\right) = V_{in}$$

To generate single-ended output, simply discard 1 output of differential pair

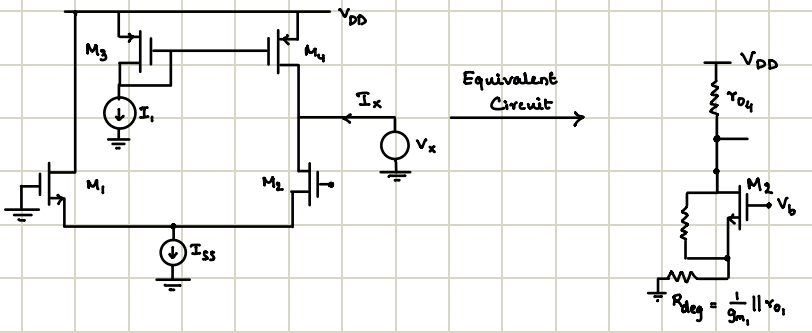
Current source in passive mirror serves as load

we know, $A_v = \frac{V_{out}}{V_{in}} = -G_M R_{out}$

and to find $G_M = \frac{V_{out}}{V_{in} \times R_{out}} = \frac{I_{out}}{V_{in}} = \frac{g_{m_1} V_{in}/2}{V_{in}} = \frac{g_{m_1}}{2}$

→ Half of I_D is wasted so g_m is reduced to half

→ For R_{out}, Short input to zero & apply test voltage at output node



$R_{out} \simeq 2r_{o_2} \,||\, r_{o_4}$

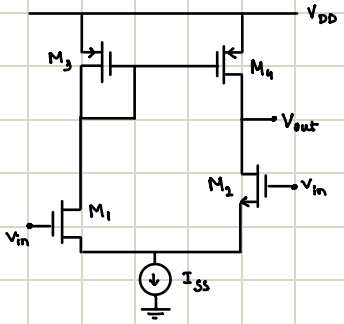
$A_v = -G_M R_{out}$

$= -\left(\frac{g_m}{2}\right)\left(2r_{o_2} \,||\, r_{o_4}\right)$

If $r_{o_4} \to \infty$ then $A_v \to -g_{m_1} r_{o_2}$

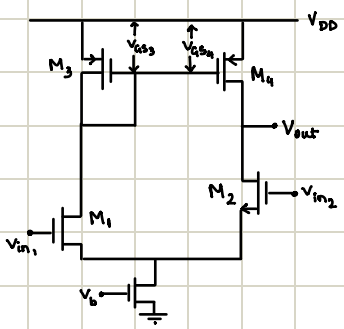
Differential Pair with Active Load (Active Current Mirror)

→ In passive load, small signal current of M_1 is wasted and A_v is halved. We want to utilize the current with proper polarity at output.



By maintaining M_3 & M_4 at same $\left(\frac{W}{L}\right)$ then we can mirror the current onto M_4 therefore not wasting current

→ Large Signal Analysis



Case 1: $V_{in_1} >> V_{in_2}$

$V_{in_1} - V_{in_2} = +ve$

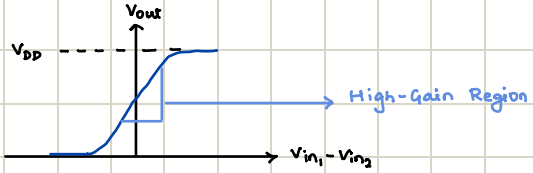
M_1 is ON, M_2 is OFF $\Rightarrow I_2 = I_4 = 0$

M_1 is conducting which means $V_{GS_3} \neq 0$, and V_{GS_4} is across same points, so, $V_{GS_4} \neq 0$

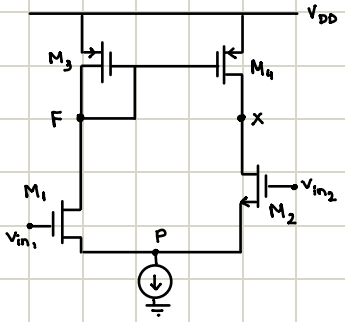
$\boxed{V_{out} = V_{DD}}$

Case 2: $V_{in_1} << V_{in_2}$

M_1 is OFF, M_2 is ON $\Rightarrow I_1 = I_3 = 0 \Rightarrow \boxed{V_{out} = 0}$



→ Small Signal Analysis



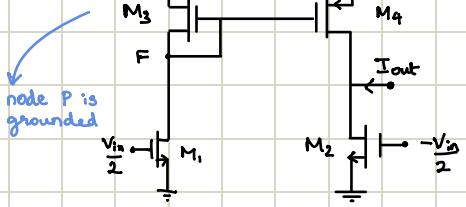
The impedances at F and X are different and so is voltage swing. This is because M_3 is diode-connected & yields lower voltage gain from input to node F. Current returning from F to P through r_{O_2} is negligible so then we can consider P as a virtual ground

Then $I_{D_1} = |I_{D_3}| = |I_{D_4}| = g_{m_{1,2}} \frac{V_{in}}{2}$

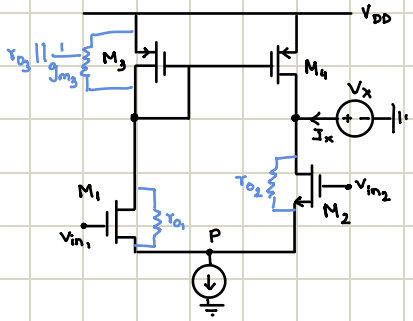
$I_{D_2} = -g_{m_{1,2}} \frac{V_{in}}{2}$

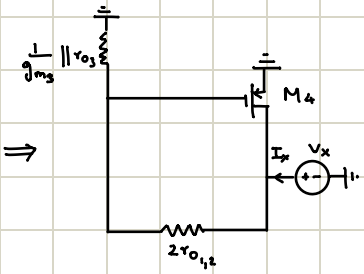
$A_V = G_M \cdot R_{out}$

$G_M = \frac{I_{out}}{V_{in}} = \frac{-g_{m_{1,2}} V_{in}}{V_{in}} = -g_{m_{1,2}}$ ⇒ Twice the value of passive load



To find Rout ⇒





Current through M_4 has 2 parts
- Mirror action from M_3 to M_4
- V_x

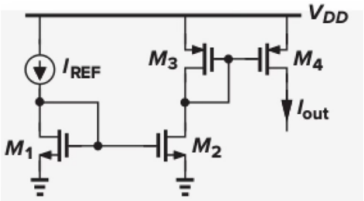
$I_x = 2I_3 + I_4$

$= \frac{2V_x}{2r_{O_2} + \left(r_{O_3} \| \frac{1}{g_{m_3}}\right)} + \frac{V_x}{r_{O_4}}$

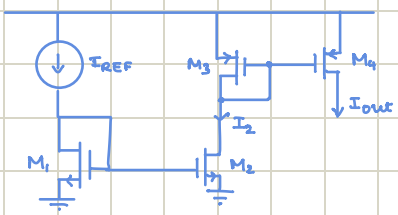
$\frac{V_x}{I_x} = \frac{1}{\frac{1}{r_{O_2}} + \frac{1}{r_{O_4}}} =$ $\boxed{r_{O_2} \| r_{O_4} = R_{out}}$

→ Which is higher than passive load

Q In the following circuit, find the drain current of M_4 if all of the transistors are in saturation.



A.

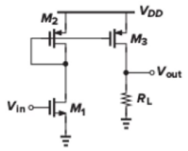


$I_2 = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} I_{REF}$

$|I_2| = |I_3|$

$I_4 = I_{out} = I_3 \times \frac{\left(\frac{W}{L}\right)_4}{\left(\frac{W}{L}\right)_3} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} \cdot \frac{\left(\frac{W}{L}\right)_4}{\left(\frac{W}{L}\right)_3} \cdot I_{REF}$

Q. Calculate the small-signal voltage gain of the circuit shown below.



A. $I_1 = g_{m_1} V_{in}$

$I_2 = I_1$

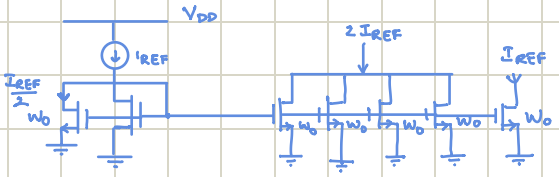
$I_3 = I_2 \cdot \frac{(W/L)_3}{(W/L)_2} = g_{m_1} V_{in} \cdot \frac{(W/L)_3}{(W/L)_2}$

$A_v = \frac{V_{out}}{V_{in}} = \frac{I_3 R_L}{V_{in}} = \frac{g_{m_1} V_{in} R_L (W/L)_3}{V_{in} (W/L)_2} = \frac{g_{m_1} R_L (W/L)_3}{(W/L)_2}$

Q. If I_{REF} is given, generate $\frac{I_{REF}}{2}$ & $2 I_{REF}$?

Assume that all transistors will have same width

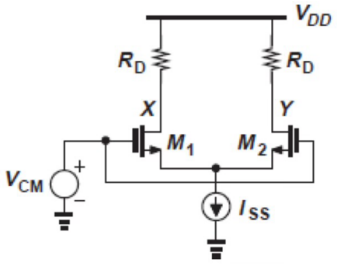
A. Initially ⇒ I_{REF}



$2 I_{REF}$ collectively because 4 of them receive $\frac{I_{REF}}{2}$.

Placing them parallelly finally adds up to give $2 I_{REF}$

Q. The MOS differential pair of Fig. must be designed for an equilibrium overdrive of 200 mV. If μnCox = 100 μA/V2 and W/L = 20/0.18, what is the required value of ISS?



A. $V_{GS} - V_{TH} = 200mV$

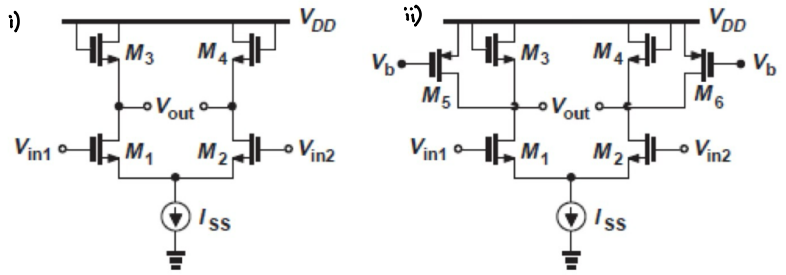
$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$

$I_D = \frac{1}{2} \times 100 \times 10^{-6} \times \frac{20}{0.18} \times (200 \times 10^{-3})^2$

$= 0.22\ mA$

$I_D = \frac{I_{SS}}{2} \Rightarrow I_{SS} = 2 I_D \Rightarrow I_{SS} = 0.44mA$

Q. Calculate the differential voltage gain of the circuits depicted in Fig. Assume perfect symmetry and $\lambda > 0$



A. i) Using half Circuit concept,

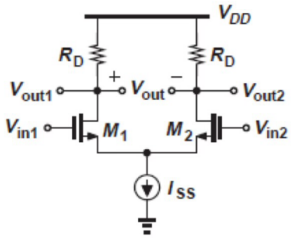
$$A_v = -g_{m_1}\left(r_{o_1} \parallel \left(r_{o_3} \parallel \frac{1}{g_{m_3}}\right)\right)$$

also parallel to $\frac{1}{g_{m_3}}$ if Gate & Drain shorted to same potential

ii) Using half Circuit concept,

$$A_v = -g_{m_1}\left(r_{o_1} \parallel \left(r_{o_3} \parallel \frac{1}{g_{m_3}}\right) \parallel r_{o_5}\right)$$

Q. Design the MOS differential pair of Fig. for an equilibrium overdrive voltage of 100 mV and a power budget of 2 mW. Select the value of RD to place the transistor at the edge of triode region for an input common-mode level of 1 V. Assume $\lambda = 0$, $\mu_n C_{ox} = 100\ \mu A/V2$, $V_{TH,n} = 0.5$ V, and $V_{DD} = 1.8$ V. What is the voltage gain of the resulting design?



A.

$$P = I_{SS} V_{DD}$$

$$I_{SS} = \frac{2\times10^{-3}}{1.8} = 1.11\ mA$$

$$V_{GS} - V_{TH} = 100\ mV$$

$$I_D = \frac{I_{SS}}{2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$

$$\frac{1.11\times10^{-3}}{100\times10^{-6}\times(100\times10^{-3})^2} = \frac{W}{L} \Rightarrow \frac{W}{L} = 1111.11$$

$$g_m = \sqrt{\mu_n C_{ox}\frac{W}{L} I_{SS}} = \sqrt{100\times10^{-6}\times1111.11\times1.1\times10^{-3}} = 0.011\ S$$

To place transistor at edge of triode region

$$V_{DS} = V_{GS} - V_{TH}$$

and $V_{out} = V_{in,CM} - V_{TH}$

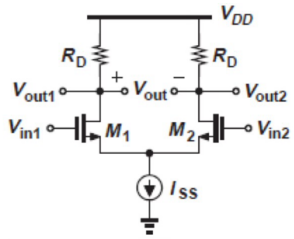
$$V_{DD} - R_D I_D = V_{in,CM} - V_{TH} \quad (V_{out} = V_{DD} - I_D R_D)$$

$$V_{DD} - \frac{R_D I_{SS}}{2} = V_{in,CM} - V_{TH}$$

$$R_D = \frac{(1.8 - 1 + 0.5)\times 2}{1.11\times10^{-3}} \Rightarrow R_D = 2.34\ k\Omega$$

$$A_v = -g_m R_D = -0.011\times2.34\times10^3 = -25.76$$

Q. Design the MOS differential pair of Fig. for a voltage gain of 5 and apower dissipation of 1 mW if the equilibrium overdrive must be at least 150 mV. Assume λ = 0, μnCox = 100 μA/V2, and VDD = 1.8V



A.

$P = 1mW$; $A_v = 5$

$V_{GS} - V_{TH} \geq 150\,mV$

$\mu_n C_{ox} = 100\,\frac{\mu A}{V^2}$; $V_{DD} = 1.8V$

$P = I_{SS} V_D \Rightarrow I_{SS} = \frac{1m}{1.8} = 0.55\,mA$

$A_v = -g_m R_D$

$g_m = \frac{2 I_D}{(V_{GS} - V_{TH})} = \frac{I_{SS}}{V_{GS} - V_{TH}} = \frac{0.55m}{150m} = 3.67\,mS$

$R = \frac{5}{3.67m} = 1.363\,K\Omega$

$I_{SS} = \mu_n C_{ox} \left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2 \Rightarrow \left(\frac{W}{L}\right)_1 = \frac{0.55 \times 10^{-3}}{100 \times 10^{-6} \times (150 \times 10^{-3})^2} = 244.45$

$\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 244.45$